# ABELIAN GROUPS MODULO TORSION IS THE CATEGORY OF Q-VECTOR SPACES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The proof is 'the claim follows directly from Proposition proposition-quotient-by-torsion-modules' - the delivered L48292 of this lane; the identifications Z-torsion = torsion groups and Q(Z) = Q are standard mechanical specializations, content verbatim.

*Editorial note.* The label proposition-quotient-abelian-groups-by-torsion-groups names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own proposition environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Same-chapter proposition-quotient-by-torsion-modules (delivered as L48292): for a ring A with total quotient ring Q(A), Mod_A modulo torsion modules is canonically equivalent to Mod_{Q(A)}
- Abelian groups as Z-modules; torsion abelian groups = torsion Z-modules; the total quotient ring of Z is Q

*Source reference.* Original reference 0B0K: sources/stacks-examples/pinned-original-source.tex, source lines 6818–6818. Prerequisite pointer invoked at 6868; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Proposition 0.1.** *The quotient of the category of abelian groups modulo its Serre subcategory of torsion groups is the category of* $\mathbf{Q}$*-vector spaces.*

*Proof.* The claim follows directly from Proposition 0B0K. □

*Date*: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.